

Paul Cohen

Mechanical Design Engineer

Portfolio

Stannah Uplifts homelift

A home accessibility product designed for the 6-generation family company, from initial brief to launch in 18 months. I was the Technical Lead for the project, responsible for outputs across mechanical, electrical and industrial design. It was awarded the OT Show Innovation Award and a Good Design Award in 2024.

- Ran early-stage architectural concept generation and refinement, covering the drive system, fail-safe brake, structural design, and safety systems.
- Managed subsystem prototyping and testing to de-risk critical systems at the earliest stage, including a novel fail-safe braking system.
- Led the internal interdisciplinary team over 12 month engagement (from 5 to 10 people).
- Managed the master-modelled CAD and 2D drawings for prototype and production for efficient transfer to manufacture.
- Engaged directly with suppliers (extruded aluminium, vacuum-forming, sheetmetal fabrication) for low-volume prototyping.
- Worked closely with the client to manage risks and requirements, working to an aggressive launch timeline.
- Secured ongoing projects with the client across other product lines.



Fluidic medical device

Designed for operating pressures up to 20 bar, this medical device incorporated a novel valving system in the disposable lineset to create an intuitive and efficient user experience. Its suitability for high-volume, low-cost automated production was a core requirement, with projected volumes of 5M units per annum.

- Led a 3–5 person team developing the disposable lineset, working closely with other subsystem leads to ensure smooth integration.
- Followed a user-centred process, supporting early-stage human factors studies to identify pain points and resolve them through design.
- Incorporated circular design practices by selecting co-recyclable materials throughout to support the client's sustainability goals.
- Developed a novel valve design to minimise component count, through theoretical analysis and multiple rounds of prototyping.
- Worked with CMOs and automation integrators to design for high-volume automated production, for processes including injection moulding, ultrasonic welding, tube bonding, siliconisation and inspection.
- Managed the procurement and commissioning of a suite of 10 injection mould tools for pre-pilot testing.
- Worked to ISO13485 throughout.

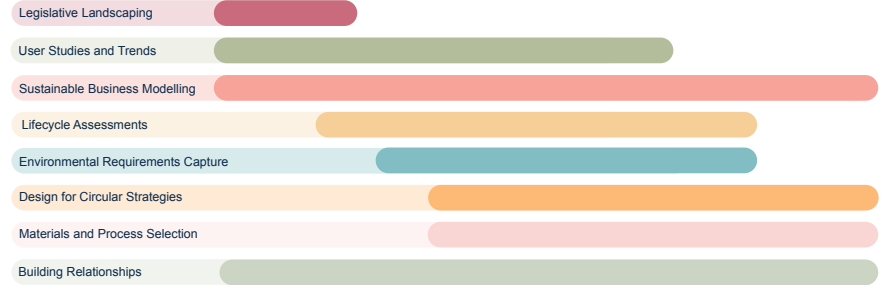


Ongoing project, restricted under NDA

KD Sustainability Lab

An employee-led initiative to incorporate sustainable and circular design practices into more projects, as well as bringing in sustainability-focused project opportunities.

- Co-founded the Sustainability Lab in 2021, creating the manifesto, strategy and internal resources for project delivery.
- Delivered workshops (internally and to clients) to educate team members and clients on topics such as Right to Repair, sustainability in healthcare, and lifecycle assessments.
- Authored sales material and opinion pieces for marketing.
- Consulted internally on sustainability aspects of projects
- Completed formal training in lifecycle assessments and circular design.



<https://www.ensera.com/company/sus-lab/>



Kypha Rapiplex

A diagnostic device for rapid point-of-care testing, offering unprecedented insight for clinicians. Developed as a platform device with multiplex testing abilities, this programme of work delivered multiple prototypes over 2 years.

- Owned and managed the overall mechanical design including precision motion system and injection moulded caseworks, moving from 1-off Alpha prototype to 13-off Beta prototypes in 12 months.
- Worked with optical engineering consultants to develop the tolerance stack through multiple lenses, filters and components.
- Developed concept architectures and component specifications based on tolerance analyses and spreadsheet calculations.
- Created and managed master-modelled CAD and 2D drawings for production.
- Designed to IEC-61010 and ISO13485 standards, including verification test design.
- Collaborated closely with electronics and software teams to ensure easy integration, hitting an accelerated delivery date of Beta prototypes with a 150-line bill-of-materials.



<https://kypha.net/products.htm#rapiplex>

Raspberry Pi 400

Raspberry Pi's first integrated computer product, the Pi 400 delivers the computing power of the Pi 4 in a compact keyboard. Portability and passive thermal performance was the goal, while still providing the flexibility that Raspberry Pi are known for.

- Conducted drop tests and identified root-cause failures on early prototypes.
- Managed production injection tool modifications with both client and supplier.
- Liaised with OEM keyboard supplier to resolve quality issues.
- Provided engineering support during production of caseworks.
- Product successfully launched in 2021.



Chargepoint CPE250 Swingarms

The CPE250 is a rapid-charging station designed to meet the demands of the growing number of electric vehicles on the road today. The swingarms sit at the top of the unit and hold the charging cables, extending the cable reach while retaining a small footprint.

- Provided onsite support for the client (Chargepoint) as part of a consulting team.
- Direct liaison with the overseas vendor, overseeing the production of 20+ unique parts, including injection-molded plastics, die-cast aluminium, and machined aluminium.
- Designed and implemented test rigs and procedures to meet lifetime engineering requirements.
- Pushed design changes and tooling modifications to meet regulatory requirements.
- Ensured parts arrived on time and in-spec for prototype milestones.
- Product successfully launched in 2019.

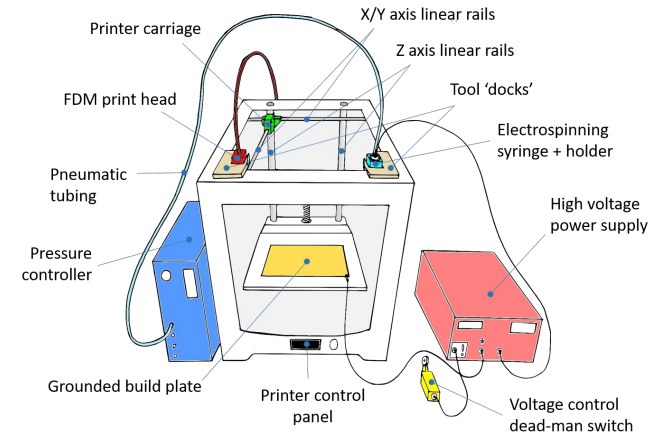


Open-source Biofabrication

Masters project. Developing an open-source biofabrication platform to produce better biological analogues for research purposes. Based on an Ultimaker 3D printer, the system requires high precision to create nanoscale fibres in a dual-process build.

- Redesigned the tool-change hardware using kinematic theory to improve the positioning accuracy
- Developed a test to validate the increased repeatability.
- Created proof-of-concept builds combining conventional 3D printing with Low-voltage Electrospinning Patterning to create suspended nano-scale fibres.

Work included for publication: <https://pubs.acs.org/doi/10.1021/acsami.9b01258>



Petra

Petra is a belaying and rappelling device for climbers and cavers, with an innovative safety feature that arrests a fall if the rope is too short. Developed as part of '2.009 – The Product Engineering Process' at MIT.

- Coordinated the team of 22 as the system integrator, chairing meetings, liaising with course staff and external suppliers.
- Led the technical design of final product, distributing jobs amongst the team and assisting with prototyping.
- Fielded technical questions during Q&A session in front of an audience of 1100.



Below: All our design iterations, with the final prototypes at the far right.

